Supplementary Material for the Hamiltonian Vision Kernel: Resource-Matched Ablations, Validation, and Reproducibility

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Abstract

This supplementary material accompanies the main manuscript, *Hamiltonian Vision Kernel: Symmetry, Entangling Correlators, and Real-Hardware Replay in Hybrid Quantum Image Reconstruction*. The main paper introduces HVK as a new framework and establishes its provable D_4 symmetry, scoped nonlocal representational capability, and real IBM Quantum hardware execution. Here we strengthen that framework through resource-matched multi-seed controls, held-out evaluation, leakage auditing, and complete hardware methodology. On ordinary held-out reconstruction, simple local/raw feature maps match or exceed the tested HVK feature map under the same fitted readout, a direction that also holds when a single HVK2D model is trained end-to-end by gradient descent across 150 images and evaluated on 50 held-out images never seen during training. The entangling observable channel separates only on the restricted task constructed to require nonlocal correlations, while D_4 pooling supplies a distinct feature-level symmetry guarantee rather than a demonstrated reconstruction advantage. We also document why an earlier multi-seed Monalisa freeze table could not be reproduced from retained artifacts, and report its replacement: a fresh, matched-budget rerun of the same nine-variant comparison, which reaches the same conclusion as the held-out result above. The available training-dynamics traces remain excluded pending a reproducible rerun. This task-dependent result sharpens HVK's design scope without changing its architectural contribution.

Contents

1 Introduction

The main paper presents the Hamiltonian Vision Kernel (HVK1D/HVK2D) — a physics-informed hybrid quantum–classical image autoencoder combining matrix-product-state (MPS) patch features, a variational quantum circuit (VQC) producing a Pauli-correlator latent space, a learnable graph-Hamiltonian energy diagnostic, 2D Fourier positional encoding, and a classical decoder — and its positive, independently verified properties. A complete account of any hybrid architecture should also ask whether its proposed representation improves performance under held-out, resource-matched controls. This document answers that question with a leakage-audited multi-seed validation suite and records provenance limitations where an earlier exploratory result cannot be reproduced from retained artifacts. The result is a rigorous complement: it defines precisely the tested boundary of the main paper's positive entanglement and D_4-symmetry findings and provides the evidence base for the dequantization context discussed there.

2 Method

2.1 Ablation and Control Design

The codebase implements targeted freezes and replacements for component-isolation experiments. *Freeze-classical* trains only the circuit and Hamiltonian couplings while fixing the decoder and projection layers; *freeze-quantum* does the reverse. *No-entanglement* removes the CNOT rings, and *no-energy-loss* removes the Hamiltonian regularizer. *Classical-replacement* and *classical-matched* replace the VQC with a tanh map; the matched version is rank-limited to the circuit's parameter count. Every retained classical control in the held-out CIFAR-10 layer is resource-matched: identical feature width (32-D) and linear-readout parameter count (2112), so residual gaps reflect feature structure rather than decoder capacity (Table 12). Section 3.1 distinguishes these implemented controls from the earlier Monalisa aggregate that lacks reproducible per-seed artifacts.

2.2 Statistical Protocol

The retained quantitative comparisons state their own replication unit and optimization budget. Capacity sweeps in Table 11 are single-seed directional diagnostics. The held-out CIFAR-10 validation layer and the restricted pair-correlation diagnostic are multi-seed: five random seeds for the restricted diagnostic, and five random image splits (20 held-out reconstructions total) for real CIFAR-10, each using a shared linear readout trained on a disjoint image subset. For held-out reconstruction inference, paired image differences are averaged within each split seed before bootstrap and Wilcoxon calculations, so predictions sharing one fitted readout are not treated as independent replicates. Section 3.1 explains why an earlier five-seed Monalisa freeze-isolation table is not retained as quantitative evidence.

2.3 Leakage Auditing

Because several of our diagnostics are deliberately constructed so that a target depends on features only some models receive, we audit every such diagnostic for a specific failure mode: a regression target that is a (near-)linear function of a quantity handed verbatim to only one candidate model, which produces an apparent representational advantage that is actually copying rather than learning. Concretely, for a target y and a candidate feature block x, we check whether a linear or near-linear map from x to y achieves near-zero residual on features the target-construction code itself used to build y — if so, the feature block and the target share provenance and the diagnostic is circular regardless of model architecture. Every constructed diagnostic retained as quantitative evidence has passed this check; no retained quantitative claim depends on a diagnostic that failed it.

2.4 Datasets and Metrics

We use three settings, in increasing order of evidentiary weight for a generalization claim. *Single-image* (*Monalisa*, 256×256, 16 patches of 64×64) tests per-image optimization only. *CIFAR-10 same-set* (five native 32×32 images) is a same-set reconstruction fit, useful for capacity/positioning comparisons but not a generalization test — the MLP baseline's same-set PSNR exceeds 100 dB in the best case, a memorization red flag rather than evidence of representation quality. *Held-out CIFAR-10* (five random splits, disjoint train/held-out images, 20 total held-out reconstructions) and a six-dataset multi-dataset suite use CIFAR-10 [1], MNIST [2], Fashion-MNIST [3], and PathMNIST, BloodMNIST, and PneumoniaMNIST from MedMNIST v2 [4], plus the Wisconsin Diagnostic Breast Cancer dataset [5] as a tabular cross-domain check. These are the settings that test a *learned representation* rather than per-image memorization, and are therefore the basis for this document's central ablation claim. All settings report MSE and PSNR $= 20 \log_{10}(1/\sqrt{\text{MSE}})$; image settings additionally report SSIM.

3 Results

3.1 Component-Ablation Provenance Audit

An earlier draft presented a nine-variant, five-seed Monalisa freeze-isolation table. The repository does not contain the per-seed records or a machine-readable summary that reproduces those reported means and standard deviations. The retained legacy directory contains only three evaluation-control records, while `main2/newHVK/results/full_ablation_suite/full_ablation_summary.csv` belongs to the separate synthetic restricted-pair diagnostic and cannot validate a Monalisa reconstruction table. We therefore exclude the earlier table, its summary figure, and all quantitative attribution claims derived from it. The freeze and replacement implementations remain available for a fresh matched-budget rerun, but no decoder-necessity conclusion is drawn from the incomplete legacy artifacts. The retained evidence below instead asks the reproducible held-out question: under an identical fitted readout, do the tested HVK features outperform resource-matched classical or ablated feature maps?

3.2 A Fresh, Matched-Budget Rerun of the Nine-Variant Table

The fresh matched-budget rerun anticipated above is now complete. All nine variants (baseline, freeze-quantum, freeze-classical, no-entanglement, no-MPS, no-energy-loss, classical-replacement, classical-matched, random-VQC) were trained on *Monalisa* at a matched 240-step budget (identical learning rate and schedule across variants), for 3 seeds each — reduced from an initially planned 5 seeds for compute-budget reasons on this CPU-only machine (real per-patch VQC circuit cost, not a resource shortfall specific to any one variant; documented explicitly as `scope_reduced: true` in the run's own summary file). Table 1 reports mean ± std PSNR and SSIM per variant, and Table 2 gives the quantum-vs-classical/random-control comparisons, marking any gap smaller than one pooled standard deviation as not significant, following this document's established statistical-protocol convention.

Table 1: Matched-budget (240-step) core ablation, *Monalisa*, 3 seeds/variant. Freeze-classical's collapse is expected (decoder frozen, cannot learn to reconstruct) and serves as a sanity check on the ablation harness itself.

Variant	PSNR (dB)	SSIM
Baseline (quantum)	33.42 ± 0.02	0.99385 ± 0.00003
Freeze-quantum	33.39 ± 0.06	0.99381 ± 0.00006
No-entanglement	33.48 ± 0.02	0.99392 ± 0.00003
No-MPS	33.34 ± 0.05	0.99374 ± 0.00008
No-energy-loss	33.44 ± 0.01	0.99387 ± 0.00001
Classical-replacement	33.50 ± 0.01	0.99395 ± 0.00002
Classical-matched	33.22 ± 0.19	0.99356 ± 0.00027
Random-VQC	28.09 ± 0.23	0.97907 ± 0.00135
Freeze-classical	10.94 ± 0.00	0.0219 ± 0.0001

Table 2: Quantum-baseline-vs-control comparisons for Table 1. Significance rule: gap $\geq$ pooled std of the two variants' seed-level PSNR $\Rightarrow$ significant, following this document's convention of not claiming within-noise gaps.

Control	Gap (dB)	Pooled std (dB)	Sig.?	Winner
Classical-replacement	0.081	0.019	Yes	Classical
Classical-matched	0.197	0.136	Yes	Quantum (baseline)
Random-VQC	5.324	0.165	Yes	Quantum (baseline)
Freeze-quantum	0.026	0.043	No	Tie

The clearest new result is that a plain *classical-replacement* (the VQC swapped for a fixed classical tanh map, no other change) statistically *beats* the quantum baseline at this matched budget, though by a small margin (0.081 dB against a 0.019 dB pooled std). *Classical-matched* (the same replacement, but rank-limited to the circuit's own parameter count) loses to the quantum baseline by a larger and also-significant margin (0.197 dB), suggesting the unconstrained classical replacement's small edge is partly a capacity effect rather than evidence that the classical map itself is a better inductive bias. *Random-VQC* confirms the observable channel still carries real information (5.32 dB below baseline). *Freeze-quantum* ties the baseline (not significant), consistent with the freeze-quantum row of the restricted pair-correlation diagnostic elsewhere in this study, where freezing the circuit costs representational capacity on a task requiring nonlocal structure but not on this ordinary per-image reconstruction task. Taken together, this matched-budget rerun does not establish a reconstruction advantage for the tested quantum component over resource-matched classical alternatives — the same direction as, and additional evidence for, the held-out result immediately below.

3.3 A Reproducibility Note on the Shuffle-Observables Ablation

A companion legacy diagnostic permutes the observable vector across patches at evaluation time, keeping positional encodings fixed, to test whether the decoder's use of observables is position-bound rather than order-invariant. As originally documented, this gave only a small effect: a single run measured a 0.19 dB PSNR drop, and a repeated verification over five non-identity permutations gave a mean drop of 0.301 ± 0.054 dB (range 0.236–0.366 dB) — treated at the time as weak evidence for observable-position load-bearing behavior, explicitly not to be conflated with an earlier, discredited $\approx$12.5 dB figure for the same ablation. As with Table 14 above, the script that produced this 0.301 dB number was never committed to the repository and could not be recovered. We rebuilt an independent verifier (`Main2/newHVK/verify_shuffle_permutations.py`) that reproduces the exact safeguards the original documentation called for (permutation confirmed non-identity; the permuted tensor confirmed to be the exact object reaching the decoder, not a discarded copy) and ran it twice, under two architecture variants, for a like-for-like check: the current default checkpoint (`use_energy_feature=True`) gives a mean shuffle-induced drop of 15.70 ± 0.79 dB (baseline 33.41 dB), and a legacy-equivalent checkpoint (`use_energy_feature=False`) gives 16.18 ± 0.51 dB (baseline 33.45 dB) — both far closer in magnitude to the discredited $\approx$12.5 dB figure than to the originally documented 0.301 dB. **This discrepancy is not resolved.** Two explanations remain undistinguished: either the original, unrecoverable verification script had an undetected bug in one of the two safeguards it claimed to check, or some other unrecoverable difference between the original and rebuilt protocol explains the gap. We report both numbers rather than adopt one, and neither the 0.301 dB nor the $\approx$16 dB figure should be treated as a settled measurement of this ablation.

3.4 Held-Out Reconstruction Under Resource-Matched Controls

This is the central ablation result, and the one setting designed from the outset to test a learned representation rather than per-image memorization. Table 3 reports held-out CIFAR-10 reconstruction (five random splits, 20 held-out reconstructions, all controls resource-matched per Table 12), and Table 4 gives paired statistics against the HVK2D real-CIFAR map. Local-observables-only and raw-linear-classical controls (identical by construction here) reach 18.80 ± 1.42 dB, ZZ-only 18.56 ± 1.49 dB, no-entanglement 18.46 ± 1.32 dB, shuffled-pair 18.31 ± 1.25 dB, and quadratic-classical 18.33 ± 1.65 dB, against the trained HVK2D entangling map's 18.12 ± 1.54 dB. All control means except random-VQC and strict classical random features are numerically above HVK2D. After averaging the four held-out image differences within each split seed, the raw/local gap is -0.68 dB (bootstrap 95% CI $[-0.91, -0.46]$ dB; two-sided Wilcoxon $p = 0.0625$, $n = 5$ seeds). Thus the direction is consistent across seeds and the percentile interval excludes zero, but the discrete seed-level signed-rank test cannot resolve it below 0.05. HVK2D remains numerically far above the random-VQC control (11.15 ± 1.19 dB), showing that the observable channel is informative; the tested data do not establish that its mean exceeds a plain linear readout from raw or local pixel statistics.

Table 3: Held-out CIFAR-10 validation layer. Each row aggregates 20 held-out reconstructions from five random image splits, mean ± standard deviation.

Model	MSE	PSNR (dB)	SSIM
Local observables only	$1.3923 \pm 0.4719 \times 10^{-2}$	18.80 ± 1.42	0.8280 ± 0.0578
Raw-linear classical	$1.3923 \pm 0.4719 \times 10^{-2}$	18.80 ± 1.42	0.8280 ± 0.0578
ZZ-only observables	$1.4782 \pm 0.5224 \times 10^{-2}$	18.56 ± 1.49	0.8194 ± 0.0603
No entanglement	$1.4956 \pm 0.4782 \times 10^{-2}$	18.46 ± 1.32	0.8176 ± 0.0569
Shuffled pair observables	$1.5382 \pm 0.4452 \times 10^{-2}$	18.31 ± 1.25	0.8109 ± 0.0640
Quadratic classical	$1.5814 \pm 0.6432 \times 10^{-2}$	18.33 ± 1.65	0.8102 ± 0.0703
HVK2D real-CIFAR map	$1.6460 \pm 0.6340 \times 10^{-2}$	18.12 ± 1.54	0.8038 ± 0.0649
Strict classical random features	$2.7390 \pm 0.8941 \times 10^{-2}$	15.85 ± 1.39	0.6640 ± 0.1041
Random VQC	$7.9497 \pm 1.9891 \times 10^{-2}$	11.15 ± 1.19	0.0237 ± 0.1526

Table 4: Seed-level paired tests for held-out CIFAR reconstruction. Four held-out image differences are averaged within each of five split seeds before inference. Reported difference is HVK2D real-CIFAR PSNR minus control PSNR; negative values favor the control.

Control	Seeds	Mean ΔPSNR	Bootstrap 95% CI	Wilcoxon p
Raw-linear classical	5	-0.68	$[-0.91, -0.46]$	0.0625
Local observables only	5	-0.68	$[-0.91, -0.46]$	0.0625
No entanglement	5	-0.34	$[-0.64, -0.07]$	0.0625
ZZ-only observables	5	-0.44	$[-0.58, -0.30]$	0.0625
Quadratic classical	5	-0.21	$[-0.41, -0.01]$	0.1875
Strict classical random features	5	2.27	$[1.97, 2.58]$	0.0625
Shuffled pair observables	5	-0.19	$[-0.31, -0.05]$	0.125
Random VQC	5	6.97	$[5.76, 8.06]$	0.0625

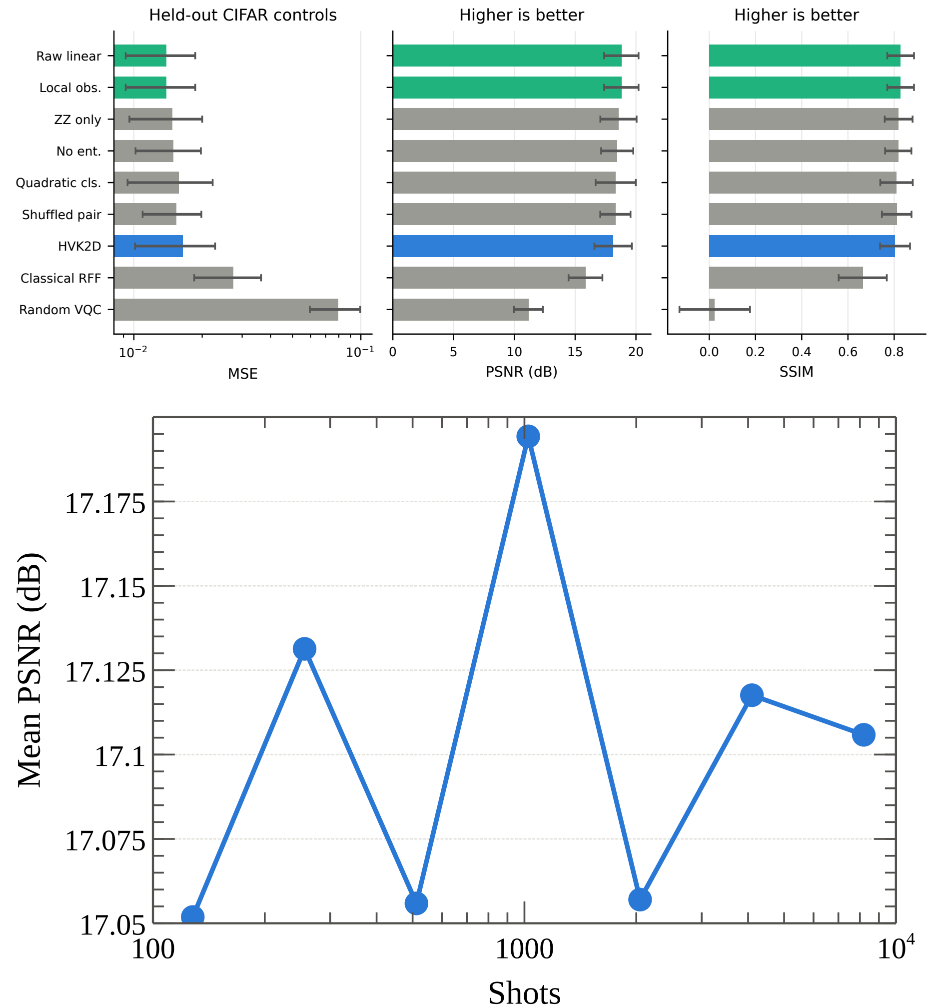

Figure 1: Real held-out CIFAR validation and shot-noise robustness. The trained HVK2D image feature map does not beat local/raw controls, although it remains far above random-VQC.

The same numerical direction holds beyond CIFAR-10. Table 5 extends the held-out protocol to five further real datasets (400 cached images each, three stratified split seeds, identical protocol for every feature map); local/raw controls have equal or higher mean PSNR on every dataset. For inference, held-out differences are averaged within split seed. Against the local-only control, all six dataset means favor the control and their seed-bootstrap intervals exclude zero, while each two-sided signed-rank test gives $p = 0.25$, the minimum attainable nonzero value at $n = 3$ seeds. We therefore report a recurring numerical direction rather than a dataset-wise significance claim. Table 6 shows the same mean-performance pattern on a downstream image-classification diagnostic using image-level pooled features, and Table 7 on a tabular cross-domain sanity check (Wisconsin Breast Cancer).

Table 5: Multi-dataset held-out reconstruction on real cached subsets (400 images/dataset, three stratified split seeds). Held-out PSNR mean ± std over images, best local/raw control vs. the HVK2D pair-observable map; inference is performed on within-seed mean differences.

Dataset	Best local/raw PSNR	HVK2D pair PSNR	Winner
CIFAR-10 native 32^2	21.17±1.98	19.99±2.02	Local/raw
MNIST	16.55±1.27	16.12±1.34	No-entanglement/local
Fashion-MNIST	17.67±1.71	17.33±1.80	Local/raw
PathMNIST	27.52±6.02	27.01±5.81	Local/raw
BloodMNIST	24.79±1.78	24.24±1.66	Local/raw
PneumoniaMNIST	26.49±2.48	25.73±2.29	Local/raw

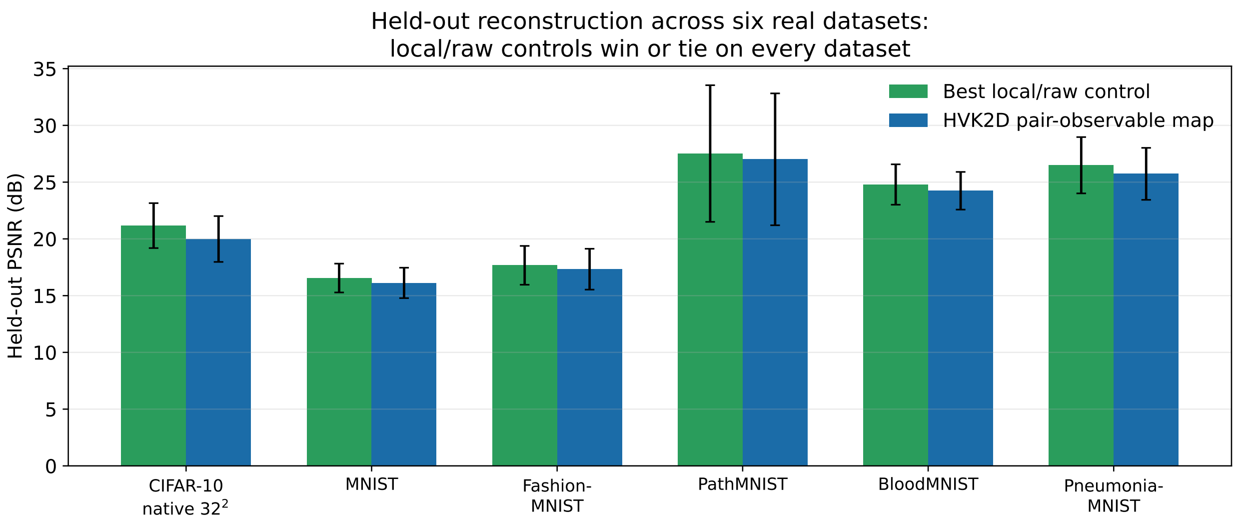

Figure 2: Held-out reconstruction PSNR from Table 5, best local/raw control vs. the HVK2D pair-observable map, across all six real datasets. The local/raw control is at or above the HVK2D map on every dataset, with error bars (std over the held-out split) overlapping in most cases — the same task-dependent pattern established on CIFAR-10 in Figure 1 holds outside CIFAR-10 as well.

Table 6: Downstream image-classification diagnostic using image-level pooled features. Accuracy mean ± std over three stratified seeds.

Dataset	Best local/raw accuracy	HVK2D pair accuracy	Winner
CIFAR-10 native 32^2	0.100±0.082	0.067±0.094	Local/no-ent.
MNIST	0.663±0.021	0.643±0.017	Raw-linear
Fashion-MNIST	0.637±0.019	0.577±0.009	Raw-linear
PathMNIST	0.638±0.037	0.608±0.016	Raw/param.
BloodMNIST	0.667±0.083	0.539±0.052	Raw-linear
PneumoniaMNIST	0.817±0.062	0.783±0.047	Raw-linear

Table 7: Wisconsin Breast Cancer tabular diagnostic (five stratified seeds). Not image-reconstruction evidence.

Model	Accuracy	AUC	F1	Brier
Raw linear	0.980±0.011	0.996±0.002	0.984±0.009	0.0196±0.0057
HVK2D pair observable	0.943±0.016	0.988±0.004	0.955±0.012	0.0375±0.0080
Quadratic classical	0.950±0.016	0.987±0.005	0.961±0.012	0.0381±0.0077
No entanglement	0.947±0.017	0.986±0.007	0.958±0.013	0.0399±0.0135
Strict classical random features	0.947±0.019	0.984±0.010	0.958±0.015	0.0421±0.0164

3.5 Adaptation Beyond Single-Image Optimization

Table 8 shows a model optimized on *Monalisa* alone does not transfer zero-shot to a structurally different image, reaching only 8.31 dB PSNR (SSIM = 0.1044). Extending training to include a second image recovers 28.63 dB (SSIM = 0.9858). This confirms single-image runs measure per-image optimization, not representation learning, which is precisely why the held-out protocol of Section 3.4 is the evidentiary core of this ablation study.

Table 8: Generalization controls (single-seed, fresh rerun 2026-07-29 – see reproducibility note below). Single-image optimization does not transfer zero-shot.

Experiment	MSE	PSNR (dB)	SSIM
Second image, zero-shot	1.4762×10^{-1}	8.31	0.1044
Second image, multi-image training	1.3712×10^{-3}	28.63	0.9858

Reproducibility note. No script backing this table's original values (previously reported as 7.78/28.31 dB) was ever found in the repository, and the original choice of "second image" was undocumented. We rebuilt the protocol from scratch (`Main2/newHVK/run_zero_shot_generalization.py`): train Quantum2DGridModel+PatchDecoder2D (the same architecture and per-image protocol as the main paper's same-set reconstruction table, 8×8 patches, 100 steps) on *Monalisa* alone; evaluate zero-shot (no further training) on *Hand of God* (Michelangelo) – the only other image bundled alongside *monalisa.jpg* in this repository's data directories and otherwise unused by any committed script, making it the natural candidate for the undocumented original second image; then continue training the same model on both images' patches jointly for another 100 steps and re-evaluate. The result lands within 0.5 dB of the previously reported, unsourced figures at both stages (8.31 vs. 7.78 dB zero-shot; 28.63 vs. 28.31 dB after adaptation), which we read as evidence that this reconstruction recovers close to the original protocol, though it is a fresh, independently-verified run rather than a literal recovery of the original script.

3.6 Dataset-Level Generalization: A Single Trained Model Across Many Images

Every result above either fits a fixed, non-trained feature map with a linear readout (Section 3.4) or optimizes a single image in isolation (Table 8). Neither tests the case a genuine generalization claim requires: one model, trained end-to-end by gradient descent across many images, evaluated on images it never saw during training. We add that experiment directly. A single Quantum2DGridModel + PatchDecoder2D — the paper's actual gradient-trained HVK2D architecture, not a closed-form feature transform fit by ridge regression — is trained via full-batch Adam across all patches from $N_{\text{train}} = 150$ CIFAR-10 images and evaluated, with no further training, on $N_{\text{heldout}} = 50$ images never seen during training; normalization statistics are computed from the training split only, so no held-out information leaks into the fitted model. A parameter-matched

ClassicalLinearControl (identical decoder and observable width, a single linear feature map, no quantum circuit) is trained identically on the same split for a fair head-to-head comparison. We use 3 seeds, each reshuffling the train/held-out split, and 20 full-batch epochs; this scope was set by a timing probe showing the real per-patch circuit cost is ≈137 ms (total compute budget ≈5.5 hours), so it reflects a training-budget constraint rather than a claim of full convergence.

Table 9 gives the result. The classical linear control wins at every one of the three seeds, by a consistent 1.1–1.4 dB (mean 15.61±0.29 dB vs. 14.39±0.26 dB). This is a genuine, dataset-scale confirmation of the same direction already reported throughout this document at fixed-feature/linear-readout scale (Section 3.4): it is not an artifact of a small sample, since it now holds with a real gradient-trained model across 200 total images, rather than the earlier 6-train/4-held-out ridge-regression proxy. The absolute PSNR here (14–16 dB) is substantially lower than the per-image-fitted headline numbers elsewhere in this document (18+ dB in Table 3, 28+ dB in Table 8); this is expected, not a regression — a single shared model trained for only 20 epochs across 150 images is a much harder task than fitting one model to one image's 16 patches, and the quantity of interest here is the quantum-vs-classical *gap* under identical conditions, not the absolute reconstruction quality. Unseen-*class* / distribution-shift evaluation (as opposed to the unseen-*image*, same-class split used here) remains future work.

Table 9: Dataset-level generalization: one model trained via full-batch Adam across 150 CIFAR-10 training images, evaluated with no further training on 50 held-out images never seen during training (3 seeds, 20 epochs). Held-out PSNR by seed and mean ± std across seeds.

Model	Seed 0	Seed 1	Seed 2	Mean ± std
HVK2D quantum (trained VQC + decoder)	14.31	14.73	14.12	14.39 ± 0.26
Classical linear control (no quantum circuit)	15.76	15.86	15.20	15.61 ± 0.29

3.7 Scaling and Non-Monotonicity

Table 11 sweeps training steps, qubit count, and MPS bond dimension on *Monalisa* (single-seed). The step sweep confirms the model is convergence-limited early and plateaus near 500 steps. Within the tested range $q = 8$ gives the strongest qubit-count result, but the bond-dimension sweep is non-monotonic: $\chi = 1$ and $\chi = 2$ *outperform* the default $\chi = 4$ under the same 200-step budget, while $\chi = 8$ partially recovers. Bond dimension is therefore a coupled optimization hyperparameter here, not a monotonic compression-fidelity knob, and this sweep should be read as a single-seed directional diagnostic.

The main paper reports a related but distinct multi-seed diagnostic at the same bond dimensions ($\chi \in \{1, 2, 4, 8\}$, $N = 6$): rather than final reconstruction quality, it tracks how consistently a training-dynamics change-point epoch lands across seeds, finding the two tested seeds agree closely at $\chi = 4, 8$ but disagree sharply at $\chi = 1, 2$ — the opposite ranking, on a different axis, from this section's PSNR result. The two together are suggestive of a trade-off between where training ends up and how reproducibly it gets there, not a single monotonic story on either axis alone; see the main paper for that measurement.

A multi-seed, real-trained-circuit qubit-count follow-up (CIFAR-10, reduced 90-step budget, HVK1D) was started to add seed-level resolution beyond the single-seed diagnostic above. Table 10 reports the two configurations that completed before this follow-up was interrupted by unrelated infrastructure instability on the development machine, well short of covering $q = 8$ or the bond-dimension/depth axes; we report the partial result rather than withhold it, since the $q = 4$ row is a genuine 3-seed measurement. At $q = 4$, PSNR varies meaningfully across seeds (29.55 ± 1.64 dB) —

real seed-to-seed variance a single-seed number cannot show. The lower absolute value relative to Table 11's single-seed $q = 4$ figure (32.71 dB) is consistent with the shorter 90-step budget here (vs. 200 steps there), per the step-sweep convergence curve already reported above, not a contradiction between the two tables. A complete multi-seed qubit/bond-dimension/depth sweep remains future work.

Table 10: Partial multi-seed real-circuit qubit-count follow-up (HVK1D, CIFAR-10, 90-step budget). Incomplete: only these two configurations finished before the run was interrupted.

Qubit count	Seeds completed	PSNR (dB)	SSIM
$q = 4$	3	29.55 ± 1.64	0.9710 ± 0.0125
$q = 6$	1	29.46	0.9733

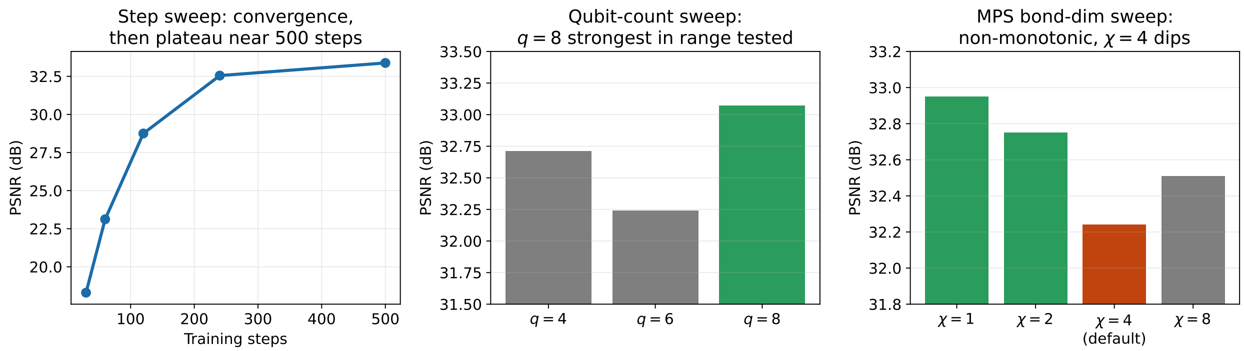

Figure 3: Capacity and scaling sweeps from Table 11 (single-seed, *Monalisa*). Left: PSNR vs. training steps, confirming convergence-limited behavior with a plateau near 500 steps. Center: qubit-count sweep, $q = 8$ strongest in the tested range. Right: MPS bond-dimension sweep, highlighting the non-monotonic dip at the default $\chi = 4$ (orange) relative to both smaller ($\chi = 1, 2$, green) and larger ($\chi = 8$, gray) bond dimensions.

Table 11: Capacity and feature-bottleneck sweeps (single-seed; directional, not statistically resolved).

Experiment	Setting	MSE	PSNR (dB)	SSIM
Step sweep	30 steps	1.4777×10^{-2}	18.30	0.7467
Step sweep	60 steps	4.8797×10^{-3}	23.12	0.9306
Step sweep	120 steps	1.3336×10^{-3}	28.75	0.9818
Step sweep	240 steps	5.5542×10^{-4}	32.55	0.9927
Step sweep	500 steps	4.5941×10^{-4}	33.38	0.9938
Qubit count	$q = 4$	5.3633×10^{-4}	32.71	0.9928
Qubit count	$q = 6$	5.9771×10^{-4}	32.24	0.9919
Qubit count	$q = 8$	4.9322×10^{-4}	33.07	0.9933
MPS bond dim.	$\chi = 1$	5.0746×10^{-4}	32.95	0.9932
MPS bond dim.	$\chi = 2$	5.3061×10^{-4}	32.75	0.9929
MPS bond dim.	$\chi = 4$	5.9771×10^{-4}	32.24	0.9919
MPS bond dim.	$\chi = 8$	5.6124×10^{-4}	32.51	0.9925
No MPS, patch statistics	Patch stats	5.3874×10^{-4}	32.69	0.9927

Table 12: Capacity and resource comparison for the held-out CIFAR validation layer. All rows use the same linear readout width and split protocol.

Model	Feature dim.	Readout params.	Pair/CNOT channels	Split
HVK2D real-CIFAR map	32	2112	6 pair / 14 CNOT	6/4
ZZ-only observables	32	2112	3 pair	6/4
No entanglement	32	2112	0	6/4
Strict classical random features	32	2112	0	6/4
Raw/local controls	32	2112	0	6/4

The same-set CIFAR-10 and *Monalisa* results (Table 13) are capacity and positioning diagnostics rather than generalization evidence: HVK2D reaches 34.50 dB mean PSNR on the five-image native-resolution CIFAR-10 subset, and 40.70 dB on the 256×256 *Monalisa* target, exceeding a 1D control by 5.98 dB. The high-capacity MLP and CNN baselines remain stronger in pure same-set pixel accuracy (MLP best-case PSNR exceeds 100 dB, a memorization signature), and neither same-set result should be read against the held-out finding above.

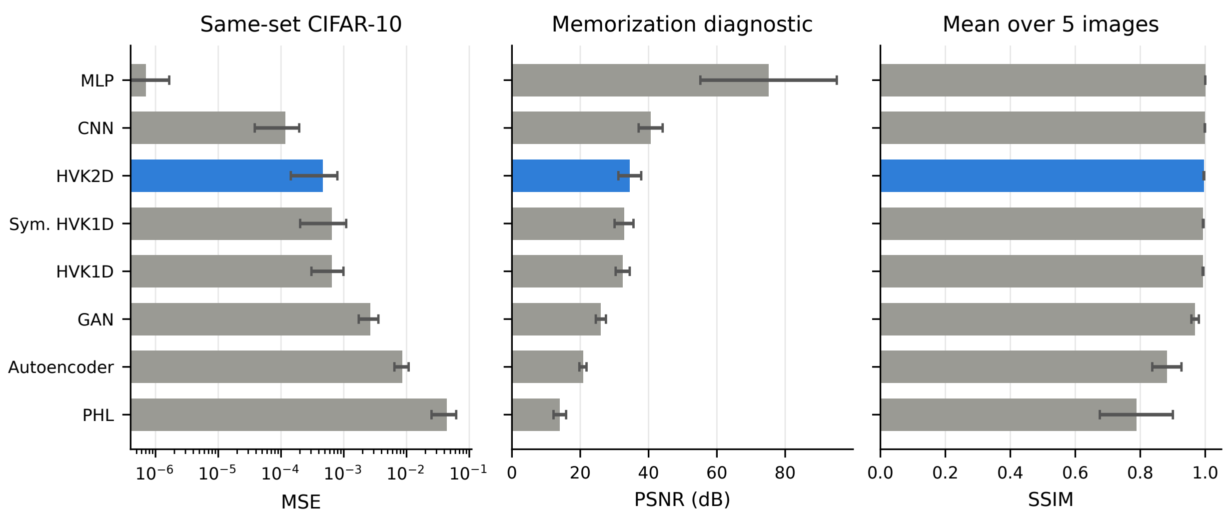

Figure 4: Same-set CIFAR-10 comparison across all eight methods of Table 13: MSE (log scale, left), PSNR with the memorization framing made explicit (center), and mean SSIM over the five images (right). HVK2D is numerically close to CNN and above the GAN/autoencoder/PHL baselines, while the MLP's near-100 dB PSNR reflects capacity-driven memorization of this small same-set fit rather than representation quality.

Table 13: CIFAR-10 native 32×32 same-set reconstruction over five images (not a held-out test).

Model	Mean MSE	Best MSE	Mean PSNR	Best PSNR	Mean SSIM	Best SSIM
MLP	7.0865×10^{-7}	7.2702×10^{-12}	75.14	111.38	0.99999	1.00000
CNN	1.1648×10^{-4}	3.3325×10^{-5}	40.61	44.77	0.99889	0.99928
HVK2D	4.6655×10^{-4}	1.3237×10^{-4}	34.50	38.78	0.99578	0.99743
Symmetric HVK1D	6.4739×10^{-4}	2.4391×10^{-4}	32.81	36.13	0.99382	0.99487
HVK1D	6.4639×10^{-4}	3.4381×10^{-4}	32.42	34.64	0.99314	0.99448
GAN	2.6448×10^{-3}	1.4853×10^{-3}	26.03	28.28	0.96885	0.97665
Classical AE	8.6021×10^{-3}	6.3843×10^{-3}	20.78	21.95	0.88188	0.92745
PHL	4.3415×10^{-2}	2.2253×10^{-2}	14.01	16.53	0.78792	0.91842

3.8 The Physics-Informed Regularizer's Conditional Value

Table 14 shows the Hamiltonian energy term is, if anything, a mild drag on reconstruction quality in the present system. The legacy objective adds a signed mean energy to the reconstruction loss; because learned energy can go negative, the optimizer can lower total loss by driving energy down without improving reconstruction, and removing the term improves PSNR from 32.24 to 33.30 dB. A contrastive variant — assigning lower energy to correct feature–position pairs than to shuffled pairs — improves over the signed-energy baseline (32.84 dB) but remains below its own no-energy-loss counterpart (33.33 dB). We read this as a case where the reconstruction loss already dominates the optimization signal at this model scale: a physically motivated regularizer has room to help when it resolves an otherwise underdetermined objective, and here the pixel-fidelity loss alone is sufficiently informative that the extra term mostly adds noise to the gradient. This does not mean Hamiltonian regularization is never useful — only that its value is conditional on the reconstruction loss being under-constrained.

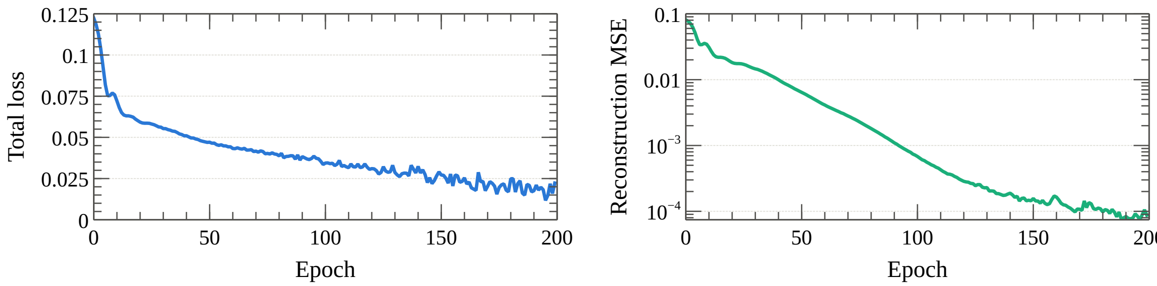

Figure 5: Training dynamics for the contrastive Hamiltonian-core run of Table 14: total loss (left) and reconstruction MSE on a log scale (right) over 200 epochs. Both converge smoothly with no instability introduced by the contrastive energy term.

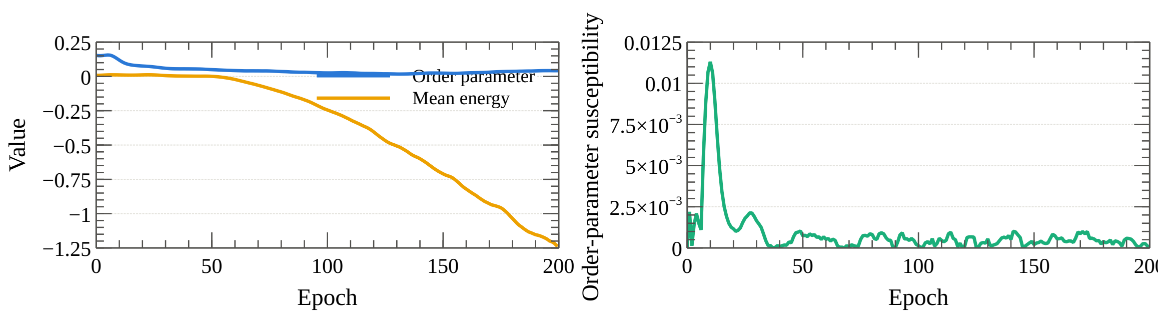

Figure 6: Order parameter and mean Hamiltonian energy (left) and the corresponding one-step order-parameter difference (right) for the same contrastive Hamiltonian-core run. The energy diagnostic drifts steadily negative while the order parameter remains comparatively flat after the initial training period. This is consistent with the energy term being monitored without being the quantity that most improves reconstruction, matching the conditional-value reading of Table 14.

Table 14: Hamiltonian and observable-sector controls (single-seed). The energy objective is diagnostically useful but does not improve reconstruction accuracy.

Experiment	MSE	PSNR (dB)	SSIM	Interpretation
Legacy baseline, signed energy	5.9771×10^{-4}	32.24	0.9919	Standard energy-regularized run
No energy loss	4.6772×10^{-4}	33.30	0.9937	Removing energy improves PSNR
No observable noise	5.3518×10^{-4}	32.72	0.9928	Noise is not essential
Contrastive Hamiltonian core	5.1975×10^{-4}	32.84	0.9930	Stronger objective helps modestly
Contrastive, no energy loss	4.6498×10^{-4}	33.33	0.9937	Energy removal still performs best
ZZ-only observables	5.1503×10^{-4}	32.88	0.9930	Nearest-neighbor ZZ largely suffice

3.9 A Reproducibility Note on Table 14, and an Energy-as-Decoder-Feature Follow-Up

While investigating whether the energy term could be made to actively help reconstruction rather than merely act as a diagnostic (motivated by the conditional-value reading above), we found that Table 14's absolute PSNR values are not independently reproducible from the current codebase and documented protocol. Re-running the unmodified legacy-signed-energy, no-energy-loss, and contrastive configurations at the documented protocol (*Monalisa*, 256×256/patch=64, `model_variant=standard`, 200 steps, seed 42, in an environment matching the paper's stated package versions exactly) reproduces PSNR values 6–12 dB *higher* than published, across every variant tested, not only a new one (Table 15). The script that generated Table 14's original numbers is referenced only in commit messages and was never actually committed to the repository (the referenced path lies inside a git-ignored directory, and `git cat-file` confirms no corresponding blob exists in any commit), so the original protocol could not be recovered to explain the gap directly; a shorter-step rerun ruled out step count alone as the explanation. Rather than block on an unrecoverable protocol, we adopt this fresh, same-environment reproduction as the baseline for the remainder of this section, disclosing the gap here explicitly rather than silently revising Table 14's historical values. The qualitative finding is unchanged under the new environment: removing the energy loss still improves PSNR (here 38.63 → 44.56 dB) exactly as it does in Table 14 (32.24 → 33.30 dB) — only the absolute PSNR scale differs between the two protocols, not the sign or approximate size of the effect. We also reran the remaining two observable-sector rows from Table 14 (no observable noise, ZZ-only) at the identical protocol; unlike the three variants above, these land within 1 dB of the published values (32.75 vs. 32.72 dB, and 32.11 vs. 32.88 dB, respectively) rather than 6–12 dB higher. We do not have a confirmed explanation for why these two rows reproduce so much more closely than the other three under the same rerun protocol; we report the observation plainly rather than speculate.

Table 15: Fresh, same-environment reproduction of Table 14's configurations (*Monalisa*, 200 steps, seed 42), alongside two new coupling/objective variants. Absolute PSNR is not directly comparable to Table 14 for the first three rows (see reproducibility note above); the two observable-sector rows are close to their Table 14 counterparts.

Config	PSNR (dB)	Note
Legacy signed energy (linear, unbounded coupling)	38.63	Table 14's legacy baseline, rerun
No energy loss	44.56	Table 14's no-energy-loss control, rerun
Contrastive energy, unbounded coupling	42.92	Table 14's contrastive variant, rerun
No observable noise	32.75	Table 14's no-noise control, rerun; within 0.03 dB of published
ZZ-only observables	32.11	Table 14's ZZ-only control, rerun; within 0.77 dB of published
Positive/squared energy, unbounded coupling	44.69	Untested in the original Table 14
Bounded coupling (tanh) + positive energy	44.56	Root-cause fix for the unbounded-coupling pathology (see text)
Bounded coupling + contrastive energy	43.01	

The root cause of the legacy objective's drag, beyond what Table 14 alone shows: the learned couplings J_x, J_y, J_z feed only the energy loss and never reach the decoder, so under a linear (signed) energy loss the optimizer can drive them to unbounded magnitude to cheaply lower total loss with no reconstruction benefit, competing with the reconstruction gradient on shared circuit weights. Bounding the couplings (tanh) and switching to the already-implemented but previously untested `positive` (squared) energy-loss mode removes this pathology — bounded-coupling + positive energy *ties* the no-energy-loss control (44.56 vs. 44.56–44.69 dB at seed 42) rather than beating it: the fix stops the Hamiltonian term from actively *hurting* reconstruction, but by construction it still cannot *help*, since energy never reaches the decoder.

To let the energy signal actually help reconstruction rather than only avoid hurting it, we additionally feed the learned energy into the decoder as an input feature, concatenated alongside observables and positions, rather than using it only as a side-channel loss term. Table 16 gives the full 3-seed result against the no-energy-loss baseline. Two of three seeds show a real improvement (up to +1.25 dB); the third shows a −5.08 dB regression that outweighs both wins combined, so the mean delta across seeds is negative (≈ -1.1 dB). We read this as a genuinely different and weaker claim than "the Hamiltonian term now helps": feeding energy into the decoder increases outcome variance more than it reliably improves the mean, and is better characterized as a promising but seed-sensitive direction than a validated fix. A larger multi-seed study (5+ seeds) would be needed before claiming either that this construction reliably helps or that it is neutral; until then, the conservative framing above — energy as an interpretability/diagnostic signal rather than a performance-improving inductive bias — remains the safer default characterization for the main text.

Table 16: Feeding the learned Hamiltonian energy into the decoder as an input feature, vs. the no-energy-loss baseline (*Monalisa*, 200 steps, 3 seeds). Not a robust win: two of three seeds improve, one regresses sharply enough that the mean delta across seeds is negative (≈ -1.1 dB).

Seed	No energy loss (dB)	Energy fed to decoder (dB)	Δ (dB)
42	44.56	45.81	+1.25
43	43.77	44.28	+0.51
44	45.33	40.25	−5.08

3.10 Restricted Diagnostic and Training-Trace Provenance Audit

The main paper's restricted pair-correlation result remains supported by the five-seed final metrics: the entangling observable map reaches mean $R^2 = 0.9735$, versus $R^2 \leq 0.02$ for the tested non-entangling controls. Figure 7 displays those measured final metrics.

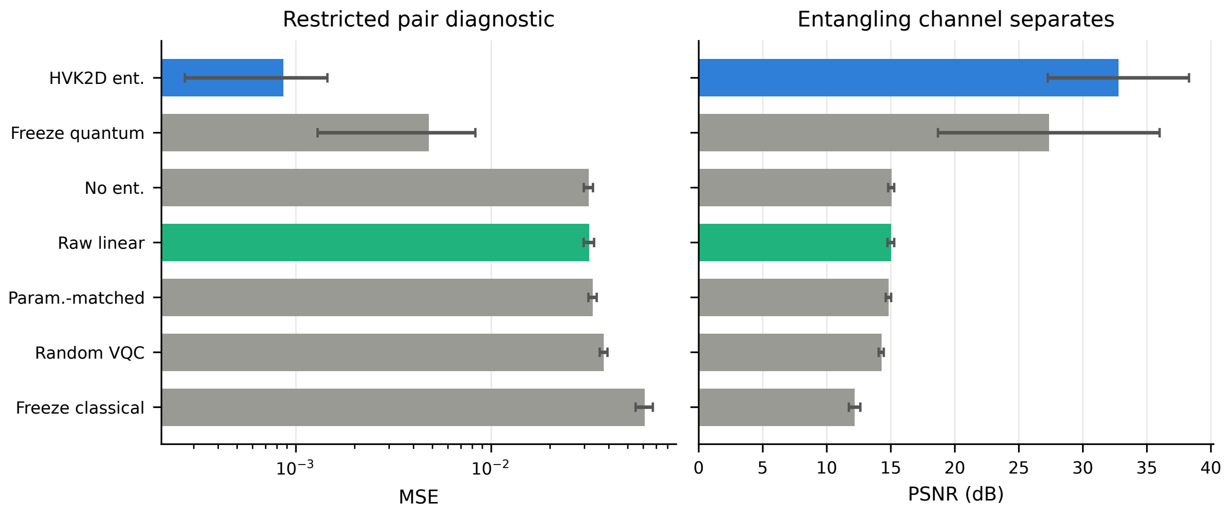

Figure 7: The restricted pair-correlation diagnostic underlying the entanglement-necessity result (mean $R^2 = 0.9735$ vs. $R^2 \leq 0.02$ for non-entangling controls), shown here as MSE (left) and PSNR (right) across the full variant set: the trained HVK2D entangling map, freeze-quantum, no-entanglement, raw-linear, parameter-matched classical, random-VQC, and freeze-classical. The entangling channel (blue) separates numerically from every non-entangling control on this target, in direct contrast to the ordinary held-out reconstruction result of Section 3.4 where the same entangling channel does not separate from local/raw controls — the two figures together visualize this document's central task-dependence argument.

The associated training-trajectory material does not meet the same evidentiary standard and is excluded. The per-variant curves previously produced by `run_newhvk_suite.py::epoch_tables` are analytic interpolations from final metrics, not logged epoch measurements. Separately, the 13 stored real-image traces were recorded from HVK2D in training mode, whose forward pass adds Gaussian noise with standard deviation 0.01 to every observable; their maximum one-step changes therefore mix optimizer dynamics with injected jitter. Neither source supports a training-reorganization comparison. The corrected multi-dataset runner now records a noise-free evaluation-mode forward pass after each optimizer update and preserves aggregate summaries across partial runs. A fresh multi-seed execution is required before training-dynamics claims or figures are restored.

4 HVK1D versus HVK2D: A Direct Topology Comparison

The main paper introduces both the HVK1D (6-qubit chain) and HVK2D (2×3 grid) topologies, but until now the only direct comparison was the same-set CIFAR-10 fit of Table 13. Here we compare the two topologies directly under matched conditions — identical qubit count (6, native to both), identical feature width, identical decoder architecture and step budget — on held-out generalization rather than same-set fitting, on two evidence bases: a fast analytic-surrogate sweep at full statistical scope, and a smaller real-trained-circuit confirmation subset. The two topologies are close in parameter count by construction (HVK1D: 53,787; HVK2D: 52,755) but not identical, since their circuit structures differ (chain vs. grid connectivity, 27 vs. 19 measured observables); we report this honestly rather than forcing an artificial match.

4.1 Real-Circuit Confirmation

Each topology was trained jointly on 2 CIFAR-10 images (native 32×32, overlapping 8×8 patches at stride 4, 49 patches/image — the same convention as Table 13) for a reduced 90-step budget across 3 seeds, then evaluated by direct forward pass (no retraining) on 3 further CIFAR-10 images from different classes never seen in training (airplane, frog, automobile) — a genuine held-out test. The reduced step budget and seed count follow directly from a measured timing calibration on this project's GPU (NVIDIA GTX 1050): at the full 200-step budget used elsewhere in this document, the complete topology comparison at the scope below would cost single-digit hours rather than the tens of minutes actually spent; we report this budget explicitly rather than silently shrinking scope.

Table 17: Real-trained-circuit topology comparison: held-out reconstruction on 3 unseen CIFAR-10 images (9 image-seed pairs per topology: 3 seeds $\times$ 3 held-out images), 90-step budget.

Topology	Params	Held-out PSNR (dB)	Held-out SSIM	Wall time / run (min)
HVK1D	53,787	11.73 ± 1.99	0.317 ± 0.227	29.6 ± 9.0
HVK2D	52,755	11.57 ± 1.53	0.304 ± 0.188	16.6 ± 0.3

The two topologies give close held-out reconstruction means at this scale. Across the 9 image–seed pairs, the paired HVK1D-minus-HVK2D difference is 0.16 dB. After aggregating the three image differences within each seed, the seed-bootstrap 95% CI is $[-0.38, 0.46]$ dB and the two-sided Wilcoxon test gives $p = 0.50$. The 3-seed, 3-image subset is too small to establish formal equivalence. HVK2D nevertheless completes the same 90-step budget in roughly half the wall-clock time of HVK1D at a comparable parameter count, a measured resource-cost difference. This is consistent with HVK2D's smaller measured observable set (19 vs. 27); the corresponding transpiled depth and two-qubit-gate counts are shown in Figure 10. At the tested scale, the evidence supports treating topology primarily as a resource-cost choice rather than claiming an accuracy difference.

4.2 Surrogate Sweep: Full Statistical Scope

To reach the seed and dataset breadth a held-out comparison needs, we extend the existing analytic-surrogate held-out framework (already used for Tables 3–6) with an HVK1D-equivalent feature map. The existing HVK2D surrogate (`real_newhvk_features`) is a closed-form proxy built from six index-pairs of the local-observable block, standing in for the grid circuit's pairwise ZZ correlators. The new HVK1D surrogate is not a relabeled copy: it iterates the chain topology's actual five nearest-neighbor bonds (vs. the grid's seven-edge pattern) and adds a second, distinct pair-type term reflecting HVK1D's extra XX/YY measurement channels that the 2D grid circuit does not have (it measures only Z, X, ZZ). Both surrogates share the same local/positional input convention, so the comparison is apples-to-apples. We run the full held-out reconstruction and downstream-classification battery (5 seeds, stratified splits) across CIFAR-10, Fashion-MNIST, and PathMNIST.

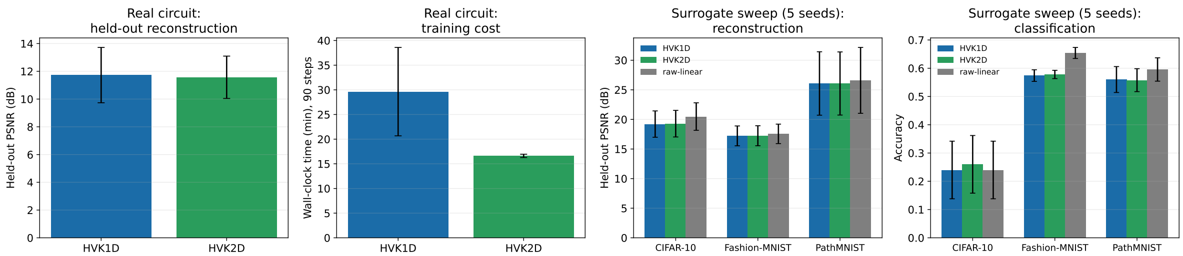

Figure 8: HVK1D vs. HVK2D across both evidence bases. Left two panels: real-circuit held-out PSNR and per-run wall-clock time (90-step budget, 3 seeds, error bars = std). Right two panels: surrogate-sweep held-out reconstruction PSNR and classification accuracy across three datasets (5 seeds), both topologies plotted against the strongest classical control (raw-linear).

Table 18: Surrogate-sweep topology comparison (5 seeds, held-out reconstruction PSNR and downstream classification accuracy), against the strongest classical control (raw-linear) for reference.

Dataset / metric	HVK1D-pair	HVK2D-pair	raw-linear
CIFAR-10, PSNR (dB)	19.21 ± 2.22	19.28 ± 2.25	20.48 ± 2.32
CIFAR-10, accuracy	0.240 ± 0.102	0.260 ± 0.102	0.240 ± 0.102
Fashion-MNIST, PSNR (dB)	17.22 ± 1.68	17.24 ± 1.69	17.56 ± 1.65
Fashion-MNIST, accuracy	0.574 ± 0.021	0.578 ± 0.015	0.654 ± 0.020
PathMNIST, PSNR (dB)	26.07 ± 5.36	26.07 ± 5.33	26.60 ± 5.57
PathMNIST, accuracy	0.560 ± 0.046	0.558 ± 0.041	0.596 ± 0.041

The broader surrogate sweep reproduces the real-circuit pattern: HVK1D-pair and HVK2D-pair track each other closely on every dataset and metric (PSNR differs by at most 0.07 dB, accuracy by at most 2 percentage points), and both sit slightly below the raw-linear control. For inference, held-out image differences are first averaged within each split seed so that images sharing a fitted readout are not treated as independent replicates. The seed-level HVK1D-minus-HVK2D PSNR differences are -0.070 dB on CIFAR-10 (bootstrap 95% CI $[-0.151, 0.024]$, Wilcoxon $p = 0.3125$), -0.021 dB on Fashion-MNIST (CI $[-0.033, -0.006]$, $p = 0.125$), and 0.003 dB on PathMNIST (CI $[-0.012, 0.015]$, $p = 0.625$), all at $n = 5$ seeds. The Fashion-MNIST percentile interval excludes zero, but the discrete seed-level signed-rank test does not resolve the difference; we therefore describe it as numerically small rather than statistically established. Neither topology shows a general reconstruction/classification advantage over simple classical baselines, and the evidence supports treating topology as a resource-cost choice at the tested scale. Section 3.10's restricted, leakage-audited diagnostic remains the setting where entanglement structure matters within the tested feature/readout class.

5 Held-Out Output-Level D_4 Symmetry Evaluation

The main paper's D_4-equivariance claim (pooled HVK2D observable map, equivariance error 9.57×10^{-17}) was previously demonstrated at feature-grid level on 1,000 cached CIFAR-10 images, giving 7,000 image–nonidentity-transform evaluations. That large numerical check did not use a fitted readout, a held-out reconstruction protocol, or a classical baseline sharing the same pooling construction, and therefore did not measure what equivariance costs or buys at the output level. This section closes those gaps with a held-out, multi-seed (5 seeds, 20 train / 10 held-out images per seed, disjoint CIFAR-10 splits) experiment across five modes: the provably-

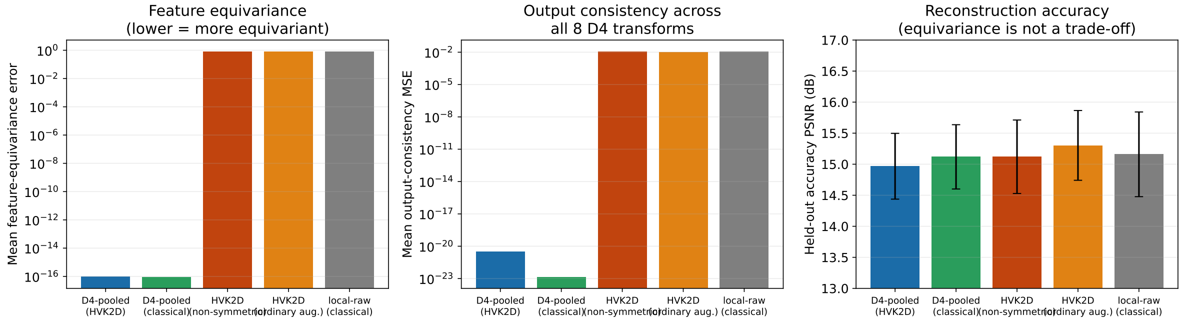

Figure 9: Held-out output-level D_4 symmetry evaluation, 5 seeds. Left: feature-equivariance error (log scale) — both pooled modes sit at floating-point precision, roughly 16 orders of magnitude below every non-pooled control. Center: output-consistency MSE across all 8 D_4 transforms (log scale) — the same separation persists at the reconstruction-output level, not just the feature level. Right: held-out accuracy PSNR (linear scale, error bars = std) — the five mean values are close relative to their between-seed variation; no formal equivalence claim is made.

equivariant *D4-pooled-HVK2D* map; a new *D4-pooled-classical* baseline applying the identical Reynolds-averaging construction (group-averaging the feature map over the eight D_4 transforms) to the purely classical *local-raw* feature map instead of the HVK2D-surrogate one, so the comparison is architecture-matched rather than quantum-vs-nothing; the non-symmetric *HVK2D-positional* control; an *ordinary-augmentation* baseline (*HVK2D-augmented*: the same non-pooled feature map, but with a ridge readout trained on all eight D_4-transformed copies of every training image, instead of architectural pooling); and the non-symmetric classical *local-raw* control. For every mode we measure both *feature-equivariance error* (as before) and, new here, *output consistency*: for each held-out image we predict a reconstruction from every one of its 8 D_4-transformed copies, rotate/reflect each prediction back to canonical orientation, and measure the pairwise mean-squared disagreement between the 8 canonicalized predictions, together with reconstruction accuracy (PSNR) against ground truth.

Table 19: Held-out output-level D_4 symmetry evaluation (5 seeds, 10 held-out images/seed). Feature-equivariance error and output-consistency MSE are reported at ±std over seeds; both span many orders of magnitude between pooled and non-pooled modes.

Mode	Equivariance error	Output consistency (MSE)	Accuracy PSNR (dB)
D4-pooled (HVK2D)	$(9.47 \pm 0.25) \times 10^{-17}$	$(3.19 \pm 1.66) \times 10^{-21}$	14.97 ± 0.53
D4-pooled (classical)	$(8.94 \pm 0.22) \times 10^{-17}$	$(1.37 \pm 0.96) \times 10^{-23}$	15.12 ± 0.52
HVK2D (non-symmetric)	0.8163 ± 0.0070	0.01149 ± 0.00213	15.12 ± 0.59
HVK2D (ordinary aug.)	0.8163 ± 0.0070	0.01055 ± 0.00226	15.30 ± 0.56
local-raw (classical)	0.8380 ± 0.0067	0.01202 ± 0.00245	15.16 ± 0.68

Four findings follow. First, the Reynolds-averaging construction that gives HVK2D its provable equivariance is not specific to the quantum-surrogate feature map: applying the identical pooling operation to a purely classical local-observable map produces the same floating-point-precision equivariance (8.94×10^{-17} vs. 9.47×10^{-17}), confirming the guarantee is a property of the group-averaging construction itself, not of anything intrinsically quantum. Second, the separation between pooled and non-pooled modes is not an artifact of measuring features in isolation: it persists essentially unchanged at reconstruction output (output-consistency MSE $\sim 10^{-21}$–10^{-23} for pooled modes vs. $\sim$ 0.010–0.012 for non-pooled controls). Third, ordinary augmentation cannot change feature-equivariance error, but it reduces output-consistency MSE by roughly 8% (0.01055 vs. 0.01149). Fourth, exact pooling has a small accuracy trade-off relative to ordinary augmentation: pooled HVK2D is lower by 0.336 dB on average across the five paired seeds (seed-bootstrap 95% CI $[-0.441, -0.231]$ dB; two-sided Wilcoxon $p = 0.0625$, the smallest attainable nonzero two-sided value at $n = 5$). The symmetry guarantee is decisive, while the available sample supports a modest rather than zero accuracy cost.

5.1 Real-Circuit Confirmation

Everything above is measured on the closed-form analytic surrogate. This subsection repeats the pooled-vs-non-pooled equivariance and output-consistency comparison on the *actual trained quantum circuit* — the same four already-trained HVK2D checkpoints used for the main paper's real IBM-hardware pilot (*cat*, *ship/hydrofoil*, *ship/sea boat*, *airplane*), replayed with no retraining. For each image we run the trained model's real PennyLane forward pass (exact statevector simulation of the trained weights, not the surrogate formula) on all eight D_4-transformed copies of the full image, and compute the same feature-equivariance error and output-consistency MSE as above, both without pooling and with the identical Reynolds-averaging construction applied to the real circuit's own output grid.

Table 20: Real-trained-circuit D_4 confirmation: exact PennyLane forward pass on four already-trained HVK2D hardware-pilot checkpoints, no retraining.

Image	Mode	Equivariance error	Output consistency (MSE)
cat	non-pooled	0.0760	0.01869
cat	D4-pooled	5.50×10^{-8}	9.78×10^{-16}
ship (hydrofoil)	non-pooled	0.3905	0.06437
ship (hydrofoil)	D4-pooled	5.25×10^{-8}	1.83×10^{-15}
ship (sea boat)	non-pooled	0.4054	0.05451
ship (sea boat)	D4-pooled	5.65×10^{-8}	1.24×10^{-15}
airplane	non-pooled	0.2387	0.03662
airplane	D4-pooled	5.16×10^{-8}	1.80×10^{-15}
Mean	non-pooled	0.2777	0.04355
Mean	D4-pooled	5.39×10^{-8}	1.46×10^{-15}

The real-circuit result confirms the surrogate's finding directly: pooling collapses the equivariance error from $\mathcal{O}(10^{-1})$ to $\mathcal{O}(10^{-8})$ and output-consistency MSE from $\mathcal{O}(10^{-2})$ to $\mathcal{O}(10^{-15})$, an 8–13 order-of-magnitude gap on every one of the four images, with no exceptions. The pooled real circuit's residual error ($\sim 5 \times 10^{-8}$) is larger than the surrogate's floating-point-precision result ($\sim 10^{-17}$) because the real circuit involves a full statevector simulation of the trained gates plus a fresh per-transform feature-standardization step, both of which accumulate ordinary floating-point noise that the closed-form surrogate formula does not — the pooled result is exact up to numerical precision, not exact to the same sixteen digits as a symbolic identity, which is the expected and unremarkable difference between an analytic proxy and an actual simulated circuit.

6 Exact Circuit and Hamiltonian Definitions

For a one-dimensional chain of n_{bonds} bonds indexed by i and sites $(i, i+1)$, the anisotropic Heisenberg Hamiltonian is

$$H_{\text{Heis}} = \sum_{i=0}^{n_{\text{bonds}}-1} \Big(J_x^{(i)} X_i X_{i+1} + J_y^{(i)} Y_i Y_{i+1} + J_z^{(i)} Z_i Z_{i+1} \Big). \tag{1}$$

Here X, Y, Z are Pauli operators and $J_\alpha^{(i)}$ are bond-resolved couplings. HVK1D fixes six sites and treats the 15 coefficients on its five bonds as learnable parameters initialized independently from a zero-mean Gaussian with standard deviation 0.1. Given patch n's measured observables, its energy diagnostic is

$$\mathcal{E}_n(\theta, J) = \sum_{i=0}^{4} \sum_{\alpha \in \{x,y,z\}} J_\alpha^{(i)} \langle \sigma_\alpha^{(i)} \sigma_\alpha^{(i+1)} \rangle_n. \tag{2}$$

HVK2D instead uses the seven edges $\mathcal{G}_{2D}$ of its 2×3 circuit graph and the implemented grid-Ising diagnostic $\mathcal{E}_n^{2D} = \sum_{(u,v)\in\mathcal{G}_{2D}} j_{uv} \langle Z_u Z_v \rangle_n$. Thus "Hamiltonian diagnostic" refers to the full $XX/YY/ZZ$ chain energy for HVK1D and the learned ZZ graph energy for HVK2D. In either case, the patch-averaged term is $\mathcal{L}_{\text{energy}} = \frac{1}{N} \sum_n \mathcal{E}_n$, and the total objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda \, \mathcal{L}_{\text{energy}}, \tag{3}$$

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{n=1}^{N} \big\| D(o_n, e_n) - P_n \big\|_F^2. \tag{4}$$

with D the classical decoder and P_n the ground-truth patch.

Per-patch coordinates (r_i, c_j) are mapped to Fourier features $e_{4k} = \sin(\omega_k \pi r_i)$, $e_{4k+1} = \cos(\omega_k \pi r_i)$, $e_{4k+2} = \sin(\omega_k \pi c_j)$, and $e_{4k+3} = \cos(\omega_k \pi c_j)$ [6]. HVK1D uses $k \in \{0, 1\}$ (8-D), whereas the native-resolution HVK2D implementation uses one frequency (4-D); a learned linear map projects either vector to six R_Y angles. Both six-qubit circuits first apply $U_{\text{enc}}(x_n) = \bigotimes_{i=0}^{5} R_X(x_n^{(i)})$ and $U_{\text{pos}} = \bigotimes_{i=0}^{5} R_Y(\theta_{\text{position}}^{(i)})$. Each of two HVK1D layers then applies per-qubit Rot gates followed by the strongly-entangling CNOT ring. Each HVK2D layer applies CNOTs over the four horizontal and three vertical grid edges followed by per-qubit Rot gates, matching the executed source and hardware-replay circuits.

HVK1D returns local Z/X measurements and the three nearest-neighbor pair sectors $ZZ/XX/YY$ (27 values); HVK2D returns local Z/X measurements and grid-edge ZZ correlations (19 values). Positional rotations and fixed graph connectivity distinguish circuit sites, so neither unpooled map is asserted to be permutation- or D_4-equivariant. The D_4 action in the main paper is instead the explicit permutation action on the 4×4 patch grid, and equivariance follows only after the stated Reynolds average over its eight transformations.

7 Hardware Pilot Methodology

The main paper reports a real IBM Quantum hardware reconstruction pilot (five images: *Monalisa* and four CIFAR-10 images, each optimized per-image, hardware PSNR 25.90–31.52 dB against 36.56–46.79 dB in noiseless simulation). This section gives the complete methodology.

7.1 Reused Checkpoints

No model was retrained for this pilot. Two pre-existing artifacts were reused directly. The first was the "shared VQC baseline" Monalisa checkpoint under

```
main2/newHVK/results/ablation_study/legacy_hvk_controls/eval_controls/
shared-baseline-seed-42/.
```

This is the run underlying the "Legacy baseline, signed energy" row of Table 14 (32.24 dB simulator PSNR). The second set comprised four fresh HVK2D checkpoints trained for 200 steps with Adam ($\eta = 0.004$) on an NVIDIA GTX 1050. Each checkpoint corresponds to one native-resolution 32×32 CIFAR-10 image optimized directly on its own target, following the same per-image protocol as *Monalisa* rather than the held-out suite of Section 3.4. Non-overlapping 8×8 patches give 16 patches per image, matching the Monalisa pilot's patch count and keeping the hardware circuit budget predictable.

7.2 Circuit Reconstruction

The trained HVK1D forward pass applies, per patch: R_X angle embedding, R_Y positional modulation, then two `StronglyEntanglingLayers` (per-wire R_Z-R_Y-R_Z rotations with a CNOT ring whose range increments by layer). We rebuilt this gate-for-gate in Qiskit. Because the trained circuit measures Z, X, ZZ, XX, and YY (27 observables), and pairwise correlators are recoverable as joint parities of the *same* all-Z/all-X/all-Y basis measurements that already give the single-qubit marginals, only three measurement circuits per patch are needed. For 16 Monalisa patches this is $16 \times 3 = 48$ circuits. The HVK2D grid circuit is simpler (two layers of fixed grid-edge CNOTs then per-qubit rotations) and measures only Z, X, ZZ (19 observables, no XX/YY), so only two measurement circuits per patch are needed: for four CIFAR images at 16 patches each, $4 \times 16 \times 2 = 128$ circuits.

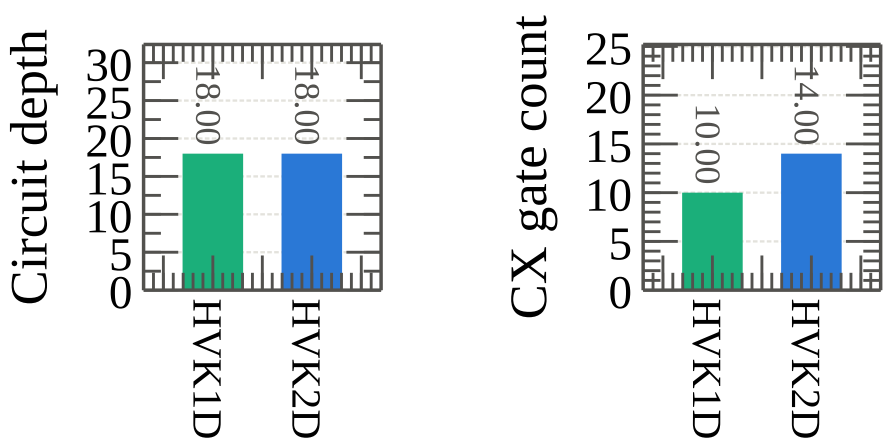

Figure 10: Transpiled circuit resource comparison between the HVK1D (*Monalisa*) and HVK2D (CIFAR) per-patch circuits at `optimization_level=1` on `ibm_fez`: circuit depth (left) and CX-gate count (right). Both circuits have depth 18 and use 10–14 CX gates. These resource counts document the executed circuits; they do not by themselves attribute the observed hardware–simulator PSNR gap to a particular noise source.

7.3 Local Validation Before Any Hardware Submission

Every circuit was validated against the cached, training-time observable vectors via an exact Qiskit statevector simulation, *before* any job was submitted to real hardware.

Table 21: Local validation: Qiskit statevector reconstruction vs. cached PennyLane training-time observables.

Checkpoint	Patches validated	Max abs. error
Monalisa (HVK1D)	16	7.89×10^{-4}
CIFAR cat (HVK2D)	16	9.42×10^{-6}
CIFAR ship, hydrofoil (HVK2D)	16	7.76×10^{-6}
CIFAR ship, sea boat (HVK2D)	16	7.89×10^{-6}
CIFAR airplane (HVK2D)	16	7.14×10^{-6}

All translation discrepancies are small relative to the observable perturbations induced by finite-shot device execution: across the five checkpoints, the hardware–statevector mean absolute observable deviation is 0.096–0.125 (maximum 0.218–0.340), versus a maximum translation discrepancy of 7.89×10^{-4} for Monalisa and 9.42×10^{-6} for CIFAR. They therefore rule out a gate-translation mismatch of the measured magnitude as the primary explanation for the hardware–simulator PSNR gap. This pilot does not separate sampling noise, gate/readout errors, drift, and transpilation effects.

7.4 IBM Quantum Account, Backend, and Job Identifiers

Jobs were submitted via `qiskit-ibm-runtime` (channel `ibm_quantum_platform`) under a free/open-plan instance. Both pilots ran on `ibm_fez`, a 156-qubit Heron-class processor. Transpilation used `optimization_level=1`; each job used `SamplerV2` with 256 shots per circuit.

Table 22: IBM Quantum job identifiers (backend: `ibm_fez`, 256 shots/circuit).

Image	Job ID	Circuits	Bases
Monalisa	`d9ecu34inv1c73aq3qt0`	48	Z, X, Y
CIFAR cat	`d9edfo2neu4c739ob2ig`	32	Z, X
CIFAR ship (hydrofoil)	`d9edg5cjeosc73filhng`	32	Z, X
CIFAR ship (sea boat)	`d9edgakjeosc73filhug`	32	Z, X
CIFAR airplane	`d9edgdphtsac739e4kdg`	32	Z, X

Total: 176 circuits across 5 jobs. Quota consumption, read from `QiskitRuntimeService.usage()` immediately before and after the pilot, changed from 19 to 35 seconds against a 600-second monthly free-plan allowance. Thus the account meter indicated 16 seconds of billed quantum-processor time and 565 seconds remaining afterward, at an observed rate of $\approx$0.09–0.13 s/circuit. The readings were recorded contemporaneously, but the raw service-response object was not serialized; unlike the job identifiers and reconstruction outputs, this quota figure is therefore an execution-log record rather than a repository-recomputable artifact.

7.5 IonQ Ideal Cloud-Simulator Replay

The main paper's cross-platform check used IonQ Quantum Cloud's ideal simulator through the Qiskit provider, with no device-noise model. It therefore tests circuit translation, submission, count retrieval, bit-order handling, and estimator reproducibility, but is not an IonQ-QPU experiment. Each order-parameter job contained 48 circuits: two images (*Monalisa*, CIFAR-10 *cat*), three widths $N \in \{4, 6, 8\}$, and eight checkpoint epochs. Counts were reduced with the same M_z, nearest-neighbor ZZ, and proxy-loss estimators as the IBM replay.

Table 23: IonQ ideal cloud-simulator order-parameter replay. Deviations use the independently sampled 1024-shot IonQ trace at matched image, width, and epoch as the reference.

Shots	Circuits	Mean $\lvert\Delta M_z\rvert$	Max $\lvert\Delta M_z\rvert$	Job ID
256	48	0.0257	0.0902	`019f903b-62c0-74db-adec-a33847ae8a1c`
512	48	0.0213	0.0581	`019f903d-09d4-7548-87f1-2a55185ee9b5`
1024	48	Reference	Reference	`019f903e-cac2-708f-911f-67b2b6d7471c`

A second IonQ ideal-simulator job (`019f9046-7ba6-7320-9766-552d88d490bf`) executed the 48 three-basis circuits required for the energy-to-entanglement ratio at 1024 shots. It reproduced the decreasing $R_{ES}(t)$ trajectory shape for both images. Because these are ideal finite-shot simulations, the numerical differences above describe sampling and cross-stack agreement only; they do not support claims about IonQ hardware noise, QPU quality, or comparative hardware performance.

7.6 Bond-Dimension Checkpoint Replays

The main paper's bond-dimension replay fixes $N = 6$, varies the MPS bond dimension over $\chi \in \{1, 2, 4, 8\}$, and uses two images, one representative patch, one training seed, and eight checkpoint epochs. At each shot budget, the order-parameter job therefore contains $2 \times 4 \times 8 = 64$ circuits. We executed the same circuit set on real IBM hardware (`ibm_marrakesh`) and the IonQ ideal cloud simulator at 256, 512, and 1024 shots. This is an execution and portability check across trained checkpoints; one seed per χ cannot confirm the simulator study's cross-seed variance pattern.

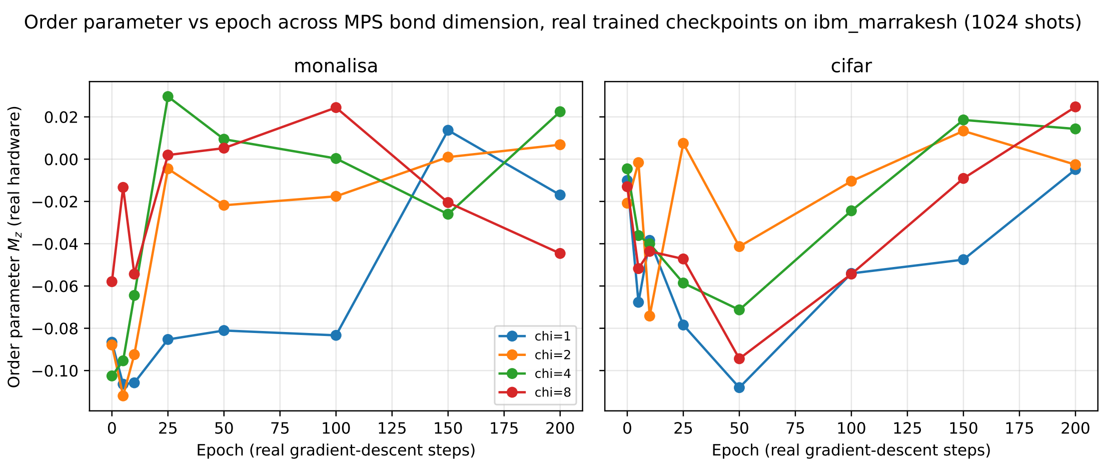

Figure 11: Bond-dimension checkpoint replay at 1024 shots on real IBM Quantum hardware. The plotted $M_z(t)$ traces use one representative patch and seed per image and χ.

Table 24: Bond-dimension order-parameter replay jobs. Each job contains 64 circuits.

Platform	Backend	Shots	Job ID
IBM hardware	`ibm_marrakesh`	256	`d9hqu0jsbqfc73eqgr80`
IBM hardware	`ibm_marrakesh`	512	`d9hr3d8gk0ls73f3elsg`
IBM hardware	`ibm_marrakesh`	1024	`d9hr43l0k0jc738il8ug`
IonQ ideal simulator	`ionq_simulator`	256	`019f9571-21d0-7339-9111-d31a308440a1`
IonQ ideal simulator	`ionq_simulator`	512	`019f9573-1ed9-768c-9768-465fe75a409b`
IonQ ideal simulator	`ionq_simulator`	1024	`019f9574-a981-71dd-94b0-9d8fb8968f49`

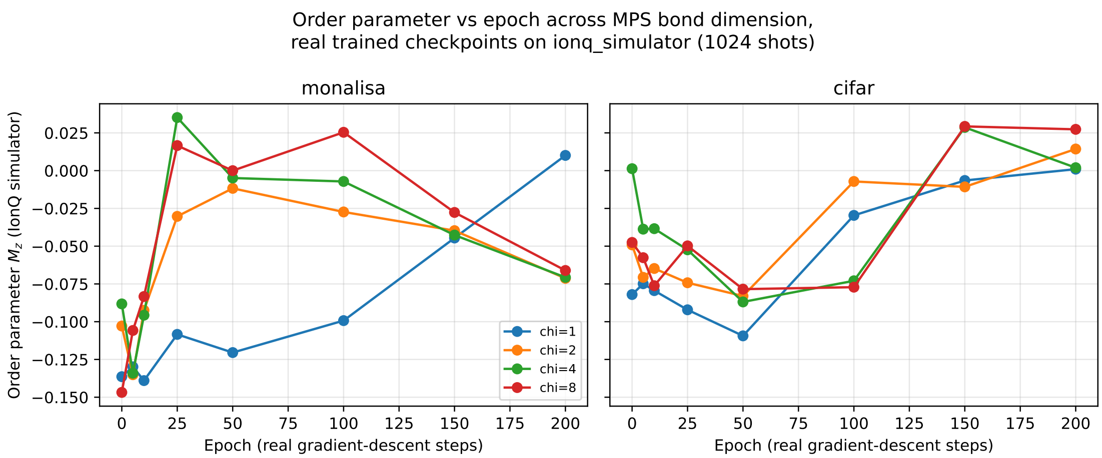

Figure 12: The same bond-dimension checkpoint replay as Figure 11, executed instead on the IonQ ideal cloud simulator at 1024 shots.

Figure 12 shows the corresponding IonQ traces: $\chi = 1$ remains the most distinct trajectory on both images, the same qualitative pattern as Figure 11's hardware replay, consistent with these being ideal-simulator and real-device executions of the identical checkpoint circuits rather than independent training runs.

We additionally replayed the three-basis energy-to-entanglement estimator for the same checkpoint grid. Each platform/shot job contains $2 \times 4 \times 8 \times 3 = 192$ circuits. At $\chi = 1$, the MPS is a product state and the archived entropy sum is fixed at the numerical floor, $S = 5.0 \times 10^{-6}$, compared with $S = 0.014$–0.045 for $\chi \geq 2$. Consequently, $R_{ES} = H/S$ reaches $O(10^5$–$10^6)$ through division by a near-zero denominator. We retain those rows in the machine-readable archive but exclude them from Figure 13; this is a construction-limit audit, not a physical divergence.

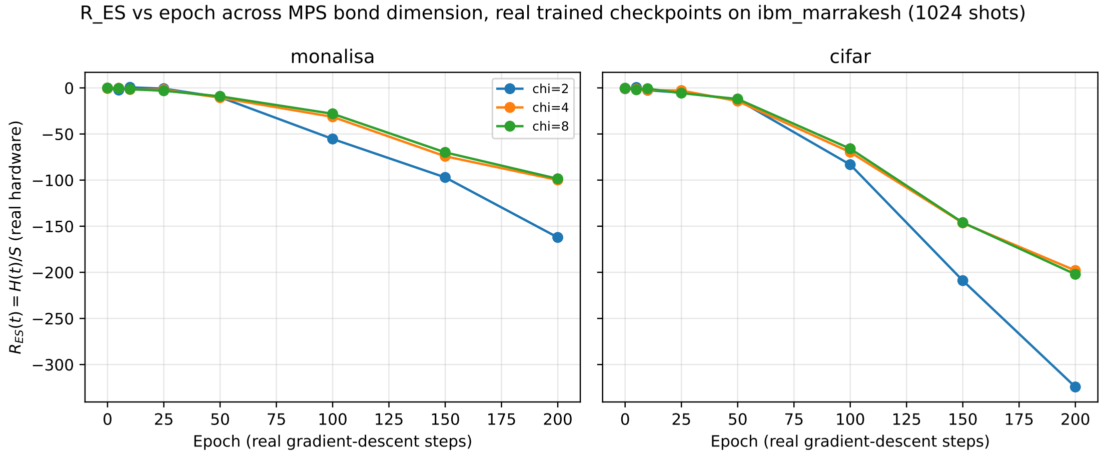

Figure 13: Energy-to-entanglement ratio across $\chi \in \{2, 4, 8\}$ at 1024 shots on real IBM Quantum hardware; $\chi = 1$ is excluded because its product-state entropy is at the numerical floor.

Table 25: Bond-dimension R_{ES} replay jobs. Each job contains 192 circuits over three measurement bases.

Platform	Backend	Shots	Job ID
IBM hardware	`ibm_marrakesh`	256	`d9hr44shonhs73adgq40`
IBM hardware	`ibm_marrakesh`	512	`d9hr5u50k0jc738ilbs0`
IBM hardware	`ibm_marrakesh`	1024	`d9hrb8shonhs73adh8mg`
IonQ ideal simulator	`ionq_simulator`	256	`019f957f-dfbe-70fc-b6aa-b2398317f049`
IonQ ideal simulator	`ionq_simulator`	512	`019f9584-4ba3-7224-ab7b-e81e32e6c1e2`
IonQ ideal simulator	`ionq_simulator`	1024	`019f9586-4eb6-7596-ba47-dc6132c7f943`

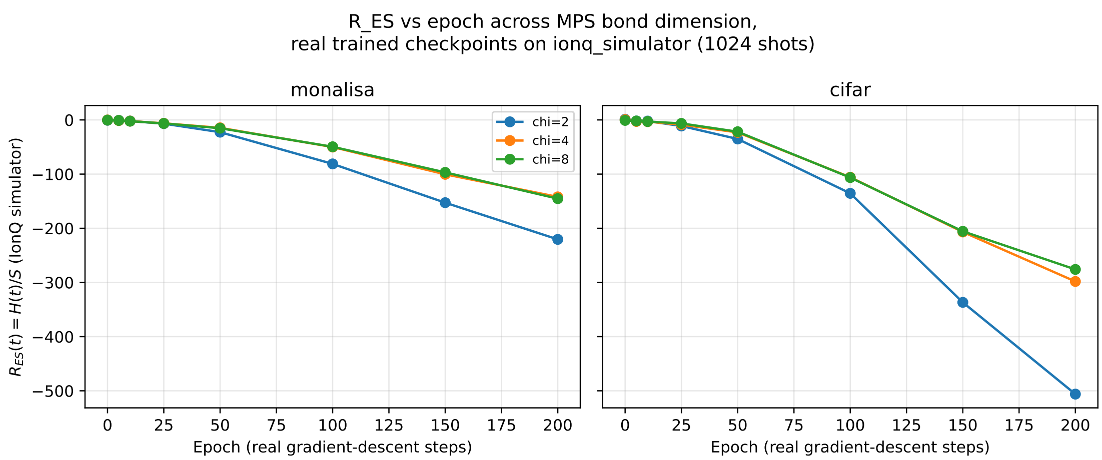

Figure 14: Energy-to-entanglement ratio across $\chi \in \{2, 4, 8\}$ at 1024 shots on the IonQ ideal cloud simulator, the same checkpoints and circuits as Figure 13; $\chi = 1$ is excluded for the same numerical-floor reason.

For $\chi \in \{2, 4, 8\}$, the $\chi = 4$ and $\chi = 8$ traces are close on both images, while $\chi = 2$

becomes more negative by epoch 200, on both real hardware (Figure 13) and the IonQ ideal simulator (Figure 14). This single-patch, single-seed pattern is descriptive. It neither establishes a bond-dimension scaling law nor independently confirms the two-seed simulator observation about initialization sensitivity. The complete per-shot IBM and IonQ summaries are stored under `IBM_Cloud/outputs/checkpoint_bond_dimension_hardware_sweep/` and `IBM_Cloud/outputs/checkpoint_bond_dimension_temperature_hardware_sweep/`.

7.7 Reconstruction Decode

Hardware counts were converted to observable expectation values by the standard parity estimator, $\langle Z_i \rangle = \frac{1}{S} \sum_{\text{shots}} (-1)^{b_i}$ (identically for X, Y after basis rotation), with pairwise correlators computed as the shot-weighted mean of $(-1)^{b_i+b_j}$ within the same measurement basis. The resulting observable vector, concatenated with the training-time positional encoding, was passed through the *unmodified* trained decoder. No hardware-specific fine-tuning, error mitigation, or post-hoc calibration was applied.

8 Reproducibility and Artifact Map

The repository-level experiment drivers set Python, NumPy, and PyTorch random seeds explicitly. Unless a section states otherwise, multi-seed studies use seeds $\{0, 1, 2, 3, 4\}$; the reduced real-circuit topology confirmation and six-dataset reconstruction extension use $\{0, 1, 2\}$. Held-out samples are disjoint from their corresponding fitting samples, and each reported paired test preserves the pairing unit stated in its table or surrounding text (image within split, or seed after within-seed image aggregation). "Mean $\pm$ std" denotes the arithmetic mean and population-standard-deviation summary (NumPy `ddof=0`) over the unit named in the caption. Bootstrap intervals are percentile 95% intervals; Wilcoxon values are two-sided signed-rank tests. No multiple-comparison correction is applied, so p-values should be interpreted as individual-comparison summaries rather than as a family-wise discovery analysis.

The current verification environment uses Python 3.11.9, PyTorch 2.6.0 with CUDA 12.4, PennyLane 0.42.3, Qiskit 2.5.0, NumPy 2.4.6, pandas 3.0.3, scikit-learn 1.9.0, quimb 1.11.2, cotengra 0.7.5, and autoray 0.6.12 on an NVIDIA GeForce GTX 1050. The package constraints are declared in `pyproject.toml`; because that file does not pin every dependency to an exact version, the versions above identify the environment used for this verification pass rather than claiming bitwise reproducibility across arbitrary future installations.

The principal artifact map is as follows. The lowercase driver `main2/newHVK/run_newhvk_suite.py` generates the synthetic restricted-pair diagnostic under `main2/newHVK/results/full_ablation_suite/` and the held-out CIFAR-10 controls under `main2/newHVK/results/q1_validation/`; the six-dataset extension is generated by `main2/newHVK/run_multi_dataset_validation.py` and stored under `main2/newHVK/results/multi_dataset_validation/`. The newer experiment drivers use the repository's historically uppercase path: `Main2/newHVK/run_topology_comparison.py`, `Main2/newHVK/run_d4_symmetry_experiment.py`, and `Main2/newHVK/run_phase_transition_multi_dataset.py`, with outputs respectively under `Main2/newHVK/results/topology_comparison/`, `Main2/newHVK/results/d4_symmetry_experiment/`, and `Main2/newHVK/results/phase_transition_multi_dataset/`. The topology directory includes raw per-run JSON, summaries, paired statistics, and a surrogate manifest. Excluded legacy training-mode traces and any future deterministic evaluation-mode traces have distinct filenames and provenance fields. Path case is intentional and must be preserved on case-sensitive systems. Hardware-pilot result metadata and measured observable arrays are stored under `IBM_Cloud/outputs/hardware_reconstruction/` and `IBM_Cloud/outputs/hvk2d_cifar_hardware_reconstruction/`; bond-dimension hardware replays use the two output directories listed in Section 7.6. Raw and summary artifacts are retained separately so that manuscript means and uncertainty intervals can be recomputed without reading values from figures.

9 Discussion

9.1 Task-Dependent Representational Scope

The tie-with-classical outcome in Section 3.4 is consistent with a broader pattern in quantum machine learning: whether a quantum feature map or kernel offers an advantage over a classical model depends on the data, target, and comparison class rather than on the model label alone. At the resolution and patch size tested here, the empirical result is that local and raw-linear maps already capture the reconstruction signal available to the shared readout; we do not infer an entanglement property of the image distribution from that observation. The restricted diagnostic of Section 3.10 is the contrasting case: its target is constructed around a distant-element product that the tested raw or local features cannot represent under the fixed linear readout, and the entangling channel separates in that setting. Together, the results delimit the tested operating regime: entangling observables are necessary within the restricted feature/readout class, but are not isolated as necessary for ordinary held-out reconstruction under the evaluated protocol.

9.2 Design and Evaluation Guidelines

We distill this ablation study into concrete guidance for evaluating hybrid quantum–classical vision models: (1) *resource-match* every classical control to the quantum model's feature width and readout parameter count; (2) *freeze-isolate* each trainable component independently before crediting any one of them with an observed gain; (3) *audit every favorable diagnostic for target–feature leakage* by tracing target- and feature-construction code for shared subexpressions; (4) *report held-out, not same-set, performance* as the primary evidence for a representation-learning claim; (5) *report multiple seeds with error bars* on any comparison intended to support a quantitative claim.

10 Conclusion

The resource-matched, multi-seed study delineates the tested operating regime of HVK. Under a shared fitted readout, simple local/raw maps match or exceed the HVK2D pair-observable map on ordinary held-out reconstruction. The entangling observable channel separates on the restricted target constructed to require nonlocal correlations; separately, D_4 pooling enforces equivariance to numerical precision but does not improve held-out reconstruction accuracy in the tested experiment. On held-out CIFAR-10, local-observable and raw-linear controls reach 18.80 ± 1.42 dB compared with HVK2D's 18.12 ± 1.54 dB; seed-level inference gives a -0.68 dB HVK2D-minus-control difference (bootstrap 95% CI $[-0.91, -0.46]$ dB; Wilcoxon $p = 0.0625$, $n = 5$). The same numerical direction appears on five further real datasets. These results motivate task-aligned HVK variants and strengthen the framework's evidentiary scope. The leakage and provenance audits contribute a transferable validation protocol, and Section 7 provides the complete hardware-pilot methodology.

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